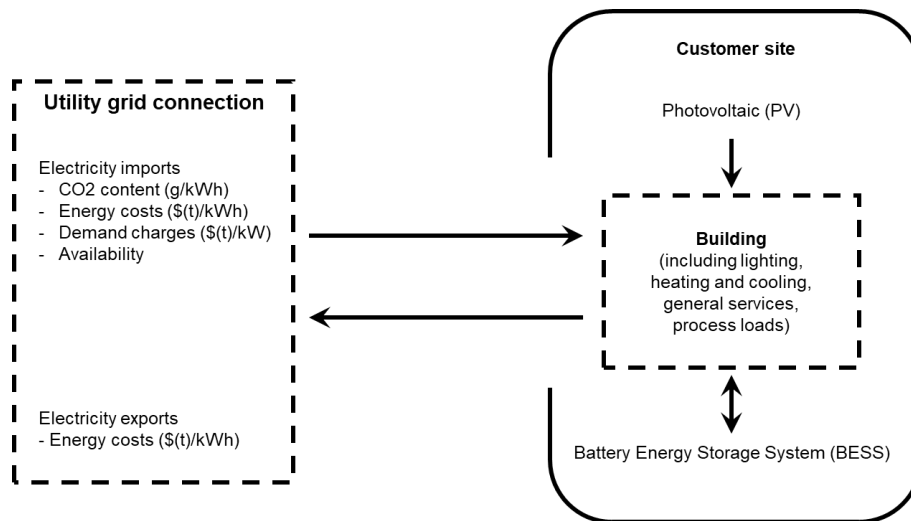


Microgrid dispatch

Motivation:

A microgrid in New Mexico, US, has been built to demonstrate economic dispatch of generation and storage assets. The microgrid consists of a utility grid connection with import and export, a photovoltaic system (PV), and a Battery Energy Storage System (BESS), all of which must be coordinated to meet the electrical load at all times. A traditional microgrid is dispatched based on a MILP optimization.



Microgrid Dispatch:

The dispatch (unit commitment) of the microgrid should minimize the costs for meeting the electrical load and maximize the potential revenue from resiliency. The State of Charge (SoC) of the BESS is the key dispatchable parameter to affect the cost of energy from the grid.

The utility grid connection has two cost components. Energy has a Time of Use (ToU) cost, with set values during defined tariff periods. Demand charges are fixed for a billing period and are applied to the highest power value that the utility grid connection draws in the billing period.

Resilience mode is when the operation of the microgrid is maintained even though the utility grid connection is unavailable. This resilience mode is provided by the BESS, supplemented by the PV. Resiliency revenue is created for every minute that the resilience mode is maintained.

Photovoltaic is a volatile energy producer. A 3-day forecast of the PV generation is available.

The microgrid load follows similar patterns of usage. A 3-day load forecast is available.

Mathematical Description:

Inputs:

- a time series of expected loads,
- a time series of expected PV generation,
- grid costs (i.e., energy and demand costs),
- BESS capacity and a simple BESS model with SoC dependent power limits,

a probability distribution for grid availability. Output: Power schedule for every time step (Grid, BESS SoC, PV, load), and resulting costs.

Optimization goal: Minimize the energy costs of microgrid for one billing period (approx. 30 days), taking resiliency revenue into account.

Optional Additions:

- Take uncertainty in (load and PV) forecasts into account
- Take into account that the real BESS State of Charge is difficult to determine in reality
- Include a non-linear BESS model